Recursive Language Model Children as Options: a Sutton-Precup-Singh correspondence

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Setup: the Sutton-Precup-Singh options framework

Recall the options framework introduced by Sutton, Precup, and Singh (1999, *Artificial Intelligence* 112). An MDP is a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$. An *option* is a triple $o = (I_o, \pi_o, \beta_o)$ where:

- $I_o \subseteq \mathcal{S}$ is the *initiation set* — states from which o can be invoked,
- $\pi_o : \mathcal{S} \to \Delta(\mathcal{A})$ is the option's *internal policy*,
- $\beta_o : \mathcal{S} \to [0, 1]$ is the *termination function* — the probability that o terminates upon reaching state s.

A high-level controller selects options in $I_o \subseteq \mathcal{S}$; each option runs to termination, after which the controller selects another option. The induced process is a *semi-Markov decision process* (SMDP).

For the SMDP, define the option-value

$$Q_\Omega(s, o) = \mathbb{E}\Big[\sum_{t=0}^{K-1} \gamma^t r_{t+1} + \gamma^K \max_{o' \in \mathcal{O}_{s_K}} Q_\Omega(s_K, o') \,\Big|\, s_0 = s,\ o_0 = o\Big],$$

where K is the (random) duration of o, r_{t+1} are intra-option rewards, and $\mathcal{O}_{s_K}$ is the set of options available at termination state s_K. SPS Theorem 1 (1999, Theorem 1) shows that SMDP Q-learning with options converges with probability 1 to Q^*_Ω under standard Robbins-Monro stepsize and ergodicity conditions.

Mapping: `rlm_query` calls as options

Recall the RLM setup of `02-math-deep-dive.md`: the agent operates in a Python REPL with primitives `FINAL`, `FINAL_VAR`, `rlm_query`. Each `rlm_query(p)` call spawns a child rollout under the same policy π_θ, returning the child's `FINAL` answer to the parent's REPL.

We propose the following correspondence. Let $\mathcal{S}_{\text{REPL}}$ be the set of REPL states (Python heap snapshots plus prompt history). Let $\mathcal{P}$ be the (possibly infinite) set of prompts that can appear as the argument of a `rlm_query` call. For each prompt $p \in \mathcal{P}$, define option $o_p = (I_p, \pi_{o_p}, \beta_p)$ by:

- $I_p = \{s \in \mathcal{S}_{\text{REPL}} : \pi_\theta \text{ may emit } \texttt{rlm\_query}(p) \text{ at } s\}$,
- $\pi_{o_p}(\cdot \mid s) = \pi_\theta(\cdot \mid s, \text{prompt} = p)$ — the same shared policy, conditioned on the child prompt p,
- $\beta_p(s) = \mathbf{1}[s \text{ contains a } \texttt{FINAL}(.) \text{ invocation}]$.

Primitive actions (one-token emissions that are not `rlm_query` calls) are kept as-is: each token-emission action is a degenerate option with $\beta = 1$ everywhere and duration $K = 1$. This makes the entire RLM session an SMDP over the high-level alphabet $\mathcal{A}_\Omega = \mathcal{A}_{\text{primitive}} \cup \{o_p : p \in \mathcal{P}\}$.

The intra-option discounted reward of o_p is the discounted sum of (zero) intra-child rewards plus the discounted child *advantage* accumulated through advantage inheritance. In the undiscounted ($\gamma = 1$) limit, $Q_\Omega(s, o_p)$ reduces to the expected root reward r_g given that the agent invoked o_p from state s and then continued under π_θ.

Theorem: convergence carry-over

Theorem 1 (SMDP Q-learning convergence for RLM SMDPs)**.** *Let θ parameterise an autoregressive policy π_θ acting in the RLM SMDP defined by the mapping above. Let $\widehat{Q}_\Omega$ be the SMDP-Q-learning estimator,*

$$\widehat{Q}_\Omega(s, o) \leftarrow \widehat{Q}_\Omega(s, o) + \alpha_n \Big[R_o + \gamma^{K_o} \max_{o' \in \mathcal{O}_{s'}} \widehat{Q}_\Omega(s', o') - \widehat{Q}_\Omega(s, o)\Big],$$

with the conditions:

(i) Robbins-Monro stepsizes: $\sum_n \alpha_n = \infty$, $\sum_n \alpha_n^2 < \infty$.

(ii) Every (s, o) pair is visited infinitely often.

(iii) Bounded reward, finite SMDP option set at each state, finite expected option duration.

Then $\widehat{Q}_\Omega \to Q^*_\Omega$ with probability 1.

Proof sketch. The RLM SMDP, under the mapping of the previous section, satisfies all hypotheses of Sutton-Precup-Singh 1999, Theorem 1: it is an SMDP with finite option set at each state (the vocabulary is finite, so $\mathcal{P}$ is countable, and only those tokens producible by π_θ at s are available); option durations are finite in expectation (the harness imposes a max number of turns per child rollout); rewards are bounded ($r_g \in [0, 1]$ for rubric scores). Conditions (i)-(iii) are precisely the SPS hypotheses. The conclusion follows by direct application of SPS Theorem 1, with no new argument required — the mapping is a faithful translation, and the SPS proof goes through unchanged. □

Proposition 2 (Hierarchical Bellman equation for RLM)**.** *For the optimal SMDP value $V^*_\Omega(s) = \max_o Q^*_\Omega(s, o)$, we have the hierarchical Bellman equation*

$$V^*_\Omega(s) = \max_{o \in \mathcal{O}_s} \mathbb{E}\big[R_o + \gamma^{K_o} V^*_\Omega(s')\big].$$

This recovers the recursive subtree loss of concept 11 as the policy-gradient analogue: the recursion in the loss is exactly the recursion in the hierarchical Bellman backup.

Proof. Standard Bellman optimality applied to the SMDP. The connection to the recursive subtree loss is by inspection: both expressions are recursive over the tree of `rlm_query` calls with the same accumulating structure. □

Discussion

What is gained from the connection.

- **Free convergence guarantee.** Theorem 1 promotes RLM-Q-learning from heuristic to convergent under standard conditions — no new proof required.

- **Hierarchical Bellman equation.** The recursive subtree loss of concept 11 is no longer just a notational convenience; it is the policy-gradient counterpart of a value-function recursion that has been studied for over two decades.
- **Options as actions for higher-level planning.** Once `rlm_query` calls are options, one can compose them into *multi-level* options (an option that invokes a sequence of `rlm_query` calls), giving a clean path to deeper hierarchical planning without changing the training algorithm.
- **Intra-option learning.** SPS 1999 introduced intra-option Q-learning, which updates Q-values from primitive transitions inside an option (not just at termination). This translates directly to RLM: every token within a child rollout could supply a Q-update for the enclosing option, accelerating learning.
- **Interruptibility.** SPS 1999 also introduced interruption: stop an option early if a higher-value option becomes available. For RLM, this is a principled way to abort a child rollout that is going badly without having to wait for its `FINAL`.

What does *not* immediately carry over. The shared-policy constraint of the paper means the *internal policies* of all options are the *same* π_θ (modulo prompt conditioning). The SPS framework allows arbitrarily different internal policies. This shared-policy restriction does not break SMDP convergence (Theorem 1 still holds) but it does mean some SPS extensions — e.g., learning option *policies* independently from the high-level controller — collapse to trivial cases.

Recommended next step. Formalise intra-option Q-learning in the RLM setting and check empirically whether it converges faster than the inheritance-only estimator of the blog. This is the most actionable corollary.